

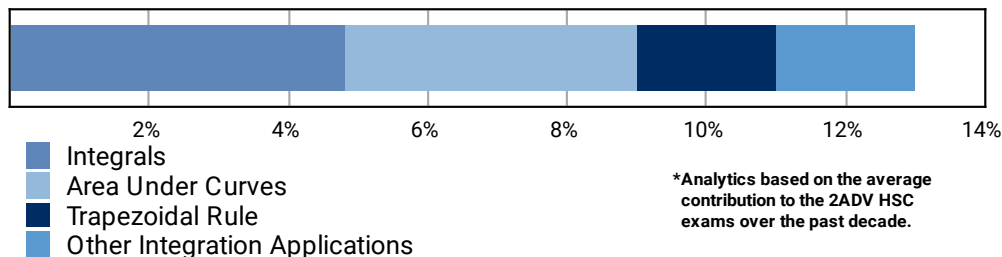
ADV: Calculus (Adv), C4 Integration (Adv), Integration (Y12)

Standard Integration

Teacher: Denis Vlismas

Exam Equivalent Time: 52.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C4 Integration



HISTORICAL CONTRIBUTION

- C4 Integration has contributed a substantial 13.0% to past Advanced exams, on average over the past decade.
- This topic has been split into four sub-topics for analysis purposes: 1-Integrals (4.8%), 2-Areas Under Curves (4.2%), 3-Trapezoidal Rule (2.0%) and 4-Other Integration Examples (2.0%).
- This analysis looks at Integrals.

HSC ANALYSIS - What to expect and common pitfalls

- Integrals (4.8%) is made up of standard integration (1.2%), log and exponential integration (2.4%) and trig integration (1.2%).
- Dedicated questions testing standard integration have appeared sporadically with a volatile mark allocation (including a full 6 mark allocation in 2009).
- Standard integrals are a great opportunity to score highly, although attention should be given to 2011 Q2e and 2011 Q4d, which produced sub-50% mean marks.
- Examined 4 times in the last 7 years, most recently in 2021.

Questions

1. Calculus, 2ADV C4 SM-Bank 4 MC

Let $f'(x) = \frac{2}{\sqrt{2x-3}}$.

If $f(6) = 4$, then

- A. $f(x) = 2\sqrt{2x-3}$
- B. $f(x) = \sqrt{2x-3} - 2$
- C. $f(x) = 2\sqrt{2x-3} - 2$
- D. $f(x) = \sqrt{2x-3} + 2$

2. Calculus, 2ADV C4 SM-Bank 5

Let $f'(x) = x^3 + x$.

Find $f(x)$ given that $f(1) = 2$. (2 marks)

3. Calculus, 2ADV C4 2016 HSC 11d

Evaluate $\int_0^1 (2x+1)^3 dx$. (2 marks)

4. Calculus, 2ADV C4 2017 HSC 11b

Find $\int (2x+1)^4 dx$. (1 mark)

5. Calculus, 2ADV C4 2009 HSC 2b

i. Find $\int 5 dx$. (1 mark)

ii. Find $\int \frac{3}{(x-6)^2} dx$. (2 marks)

iii. Evaluate $\int_1^4 x^2 + \sqrt{x} dx$. (3 marks)

6. Calculus, 2ADV C4 2007 HSC 2bii

Evaluate $\int_1^4 \frac{8}{x^2} dx$. (3 marks)

7. Calculus, 2ADV C4 2010 HSC 2e

Given that $\int_0^6 (x + k) dx = 30$, and k is a constant, find the value of k . (2 marks)

8. Calculus, 2ADV C4 2019 MET1 2

Find $f(x)$ given that $f(1) = -\frac{7}{4}$ and $f'(x) = 2x^2 - \frac{1}{4}x^{-\frac{2}{3}}$. (2 marks)

9. Calculus, 2ADV C4 2021 HSC 15

Evaluate $\int_{-2}^0 \sqrt{2x+4} dx$. (2 marks)

10. Calculus, 2ADV C4 2010 HSC 2di

Find $\int \sqrt{5x+1} dx$. (2 marks)

11. Calculus, 2ADV C4 2019 HSC 11e

Evaluate $\int_0^1 \frac{1}{(3x+2)^2} dx$. (2 marks)

12. Calculus, 2ADV C4 2014 HSC 11d

Find $\int \frac{1}{(x+3)^2} dx$. (2 marks)

13. Calculus, 2ADV C4 2011 HSC 4d

i. Differentiate $y = \sqrt{9-x^2}$ with respect to x . (2 marks)

ii. Hence, or otherwise, find $\int \frac{6x}{\sqrt{9-x^2}} dx$. (2 marks)

14. Calculus, 2ADV C4 2011 HSC 2e

Find $\int \frac{1}{3x^2} dx$. (2 marks)

Worked Solutions

1. Calculus, 2ADV C4 SM-Bank 4 MC

$$f'(x) = \frac{2}{\sqrt{2x-3}}$$

$$f(x) = 2 \int (2x-3)^{-\frac{1}{2}}$$

$$= 2 \cdot \frac{1}{2} \cdot 2(2x-3)^{\frac{1}{2}} + c$$

$$= 2\sqrt{2x-3} + c$$

When $x = 6$, $f(x) = 4$:

$$4 = 2\sqrt{12-3} + c \Rightarrow c = -2$$

$$\therefore f(x) = 2\sqrt{2x-3} - 2$$

$$\Rightarrow C$$

2. Calculus, 2ADV C4 SM-Bank 5

$$f'(x) = x^3 + x$$

$$f(x) = \int x^3 + x dx$$

$$= \frac{1}{4}x^4 + \frac{1}{2}x^2 + c$$

Given $f(1) = 2$:

$$2 = \frac{1}{4} + \frac{1}{2} + c$$

$$c = \frac{5}{4}$$

$$\therefore f(x) = \frac{1}{4}x^4 + \frac{1}{2}x^2 + \frac{5}{4}$$

3. Calculus, 2ADV C4 2016 HSC 11d

$$\begin{aligned}\int_0^1 (2x+1)^3 dx &= \frac{1}{4} \cdot \frac{1}{2} \left[(2x+1)^4 \right]_0^1 \\&= \frac{1}{8} [3^4 - 1^4] \\&= \frac{1}{8} \times 80 \\&= 10\end{aligned}$$

4. Calculus, 2ADV C4 2017 HSC 11b

$$\begin{aligned}\int (2x+1)^4 dx &= \frac{1}{5} \times \frac{1}{2} \times (2x+1)^5 + C \\&= \frac{(2x+1)^5}{10} + C\end{aligned}$$

5. Calculus, 2ADV C4 2009 HSC 2b

i. $\int 5 dx = 5x + C$

ii.
$$\begin{aligned}\int \frac{3}{(x-6)^2} dx &= 3 \int (x-6)^{-2} dx \\&= 3 \times \frac{1}{-1} \times (x-6)^{-1} + c \\&= \frac{-3}{(x-6)} + c\end{aligned}$$

iii.
$$\begin{aligned}\int_1^4 x^2 + \sqrt{x} dx &= \int_1^4 \left(x^2 + x^{\frac{1}{2}} \right) dx \\&= \left[\frac{1}{3} x^3 + \frac{2}{3} x^{\frac{3}{2}} \right]_1^4 \\&= \left[\frac{x^3}{3} + \frac{2}{3} x^{\frac{3}{2}} \right]_1^4 \\&= \left[\left(\frac{4^3}{3} + \frac{2}{3} \times 4^{\frac{3}{2}} \right) - \left(\frac{1}{3} + \frac{2}{3} \right) \right] \\&= \left[\left(\frac{64}{3} + \frac{16}{3} \right) - \frac{3}{3} \right] \\&= \left[\frac{80}{3} - \frac{3}{3} \right] \\&= \frac{77}{3}\end{aligned}$$

6. Calculus, 2ADV C4 2007 HSC 2bii

$$\begin{aligned}\int_1^4 \frac{8}{x^2} dx &= 8 \int_1^4 x^{-2} dx \\&= 8 \left[-\frac{1}{x} \right]_1^4 \\&= 8 \left[\left(-\frac{1}{4} \right) - \left(-\frac{1}{1} \right) \right] \\&= 8 \left(\frac{3}{4} \right) \\&= 6\end{aligned}$$

7. Calculus, 2ADV C4 2010 HSC 2e

$$\begin{aligned}\int_0^6 (x+k) dx &= 30 \\ \int_0^6 (x+k) dx &= \left[\frac{1}{2} x^2 + kx \right]_0^6 \\&= \left[\left(\frac{1}{2} \times 6^2 + 6 \times k \right) - 0 \right] \\&= 18 + 6k \\ \Rightarrow 18 + 6k &= 30 \\ 6k &= 12 \\ \therefore k &= 2\end{aligned}$$

8. Calculus, 2ADV C4 2019 MET1 2

$$\begin{aligned}f(x) &= \int 2x^2 - \frac{1}{4}x^{-\frac{2}{3}} dx \\&= \frac{2}{3}x^3 - \frac{1}{4} \cdot \frac{1}{\frac{1}{3}}x^{\frac{1}{3}} + c \\&= \frac{2}{3}x^3 - \frac{3}{4}x^{\frac{1}{3}} + c\end{aligned}$$

$$\text{Given } f(1) = -\frac{7}{4}:$$

$$\begin{aligned}-\frac{7}{4} &= \frac{2}{3} - \frac{3}{4} + c \\ c &= -\frac{5}{3}\end{aligned}$$

$$\therefore f(x) = \frac{2}{3}x^3 - \frac{3}{4}x^{\frac{1}{3}} - \frac{5}{3}$$

9. Calculus, 2ADV C4 2021 HSC 15

$$\begin{aligned}\int_{-2}^0 (2x+4)^{\frac{1}{2}} dx &= \left[\frac{2}{3} \cdot \frac{1}{2} (2x+4)^{\frac{3}{2}} \right]_{-2}^0 \\&= \frac{1}{3} \left(4^{\frac{3}{2}} - 0 \right) \\&= \frac{8}{3}\end{aligned}$$

10. Calculus, 2ADV C4 2010 HSC 2di

$$\begin{aligned}\int \sqrt{5x+1} dx &= \frac{1}{\frac{3}{2}} \times \frac{1}{5} \times (5x+1)^{\frac{3}{2}} + C \\&= \frac{2}{15} (5x+1)^{\frac{3}{2}} + C\end{aligned}$$

11. Calculus, 2ADV C4 2019 HSC 11e

$$\begin{aligned}
\int_0^1 \frac{1}{(3x+2)^2} dx &= \int_0^1 (3x+2)^{-2} dx \\
&= \left[-\frac{1}{3}(3x+2)^{-1} \right]_0^1 \\
&= \left[-\frac{1}{3} \cdot \frac{1}{5} - \left(-\frac{1}{3} \cdot \frac{1}{2} \right) \right] \\
&= -\frac{1}{15} + \frac{1}{6} \\
&= \frac{1}{10}
\end{aligned}$$

12. Calculus, 2ADV C4 2014 HSC 11d

$$\begin{aligned}
&\int \frac{1}{(x+3)^2} dx \\
&= \int (x+3)^{-2} dx \\
&= \frac{1}{-1} \cdot (x+3)^{-1} + C \\
&= \frac{-1}{(x+3)} + C
\end{aligned}$$

13. Calculus, 2ADV C4 2011 HSC 4d

$$\begin{aligned}
\text{i. } y &= \sqrt{9-x^2} \\
&= (9-x^2)^{\frac{1}{2}}
\end{aligned}$$

$$\begin{aligned}
\frac{dy}{dx} &= \frac{1}{2} \times (9-x^2)^{-\frac{1}{2}} \times \frac{d}{dx}(9-x^2) \\
&= \frac{1}{2} \times (9-x^2)^{-\frac{1}{2}} \times -2x \\
&= -\frac{x}{\sqrt{9-x^2}}
\end{aligned}$$

$$\begin{aligned}
\text{ii. } \int \frac{6x}{\sqrt{9-x^2}} dx &= -6 \int \frac{-x}{\sqrt{9-x^2}} dx \\
&= -6(\sqrt{9-x^2}) + C \\
&= -6\sqrt{9-x^2} + C
\end{aligned}$$

14. Calculus, 2ADV C4 2011 HSC 2e

$$\begin{aligned}
\int \frac{1}{3x^2} dx &= \frac{1}{3} \int x^{-2} dx \\
&= \frac{1}{3} \times \frac{1}{-1} \times x^{-1} + C \\
&= -\frac{1}{3x} + C
\end{aligned}$$

IMPORTANT: Some students might find calculations easier by rewriting the equation as $y = (9-x^2)^{\frac{1}{2}}$.

♦ Mean mark 46%.
MARKER'S COMMENT: Students who took the $\frac{1}{3}$ out the front before integrating made less errors, as illustrated in the Worked Solution.